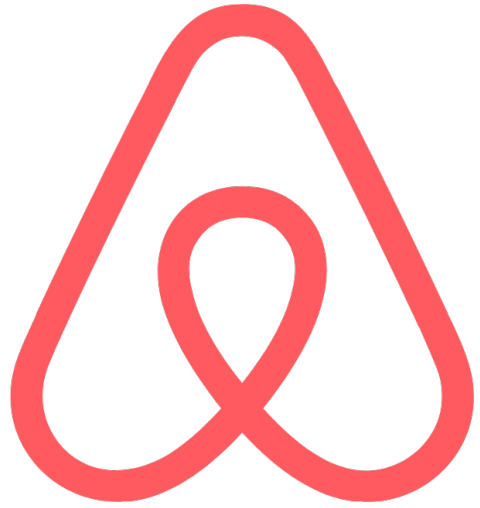




Airbnb Zurich

A Data Mining and Analysis Project on Airbnb Data





Allow me to paint a picture of Zurich...

- Largest city in Switzerland.
- It's a major global finance hub.
- Famous for its incredibly clean streets and water.
- Home to one of the world's priciest shopping streets.
- Surrounded by nature (like, really close).





Processing & Cleaning Data

Treating Missing Values



High Level Overview of Cleaning the Dataset...

Missing Data Overview

Percentage of Missing Values by Column

#	Column	% Non-Missing	#	Column	% Missing
1	license	100.00%	14	review_scores_checkin	22.14%
2	calendar_updated	100.00%	15	first_review	21.82%
3	bathrooms	100.00%	16	last_review	21.82%
4	host_neighbourhood	98.62%	17	review_scores_rating	21.82%
5	neighborhood_overview	52.64%	18	reviews_per_month	21.82%
6	neighbourhood	52.64%	19	host_response_rate	20.76%
7	host_about	39.19%	20	host_response_time	20.76%
8	bedrooms	31.37%	21	host_location	17.44%
9	review_scores_cleanliness	22.14%	22	host_acceptance_rate	14.36%
10	review_scores_accuracy	22.14%	23	description	1.78%
11	review_scores_location	22.14%	24	beds	1.74%
12	review_scores_value	22.14%	25	host_is_superhost	1.66%
13	review_scores_communication	22.14%	26	bathrooms_text	0.08%

- Treated nulls based on variable meaning (not one-size-fits-all imputation)
- Inferred missing values (e.g., bathrooms, bedrooms, beds) using related variables like accommodates
- Cleaned and encoded variables like host_response_rate and host_response_time
- Identified extreme values (e.g., unusually high beds/bedrooms) and handled before modeling
- Replaced nulls with empty strings in description and host_about for compatibility
- Dropped features with excessive missingness or low variability (e.g., license, calendar_updated)



Summary Stats

Top 10 Neighbourhoods by Listings in Zurich



Airbnb Market Insights – Zurich

Pricing, Ratings & Reviews Across Top 10 Neighborhoods



Highest Avg. Price
CHF 267.86
Rathaus



Highest Avg. Rating
4.83
Unterstrass

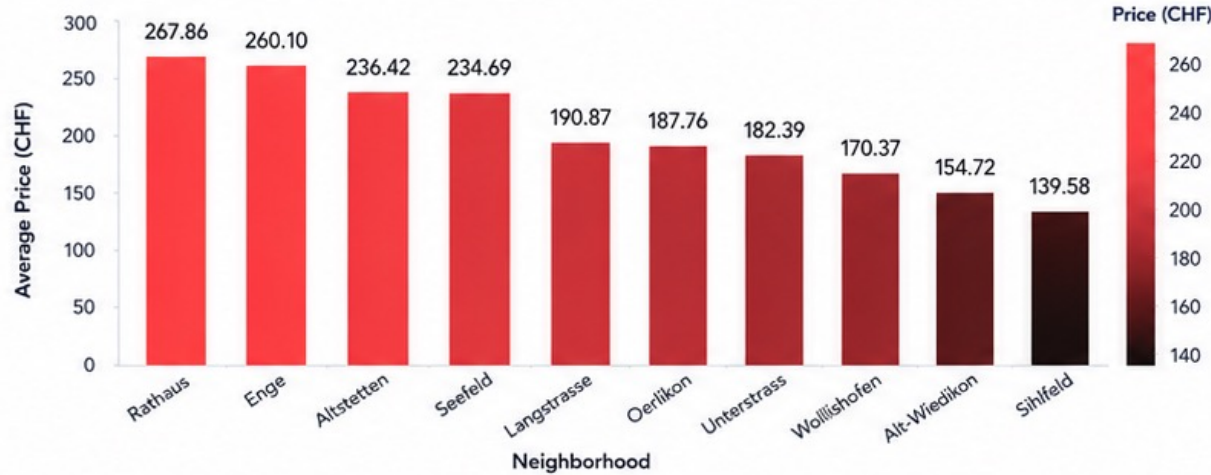


Most Reviews (Avg.)
72.95
Rathaus

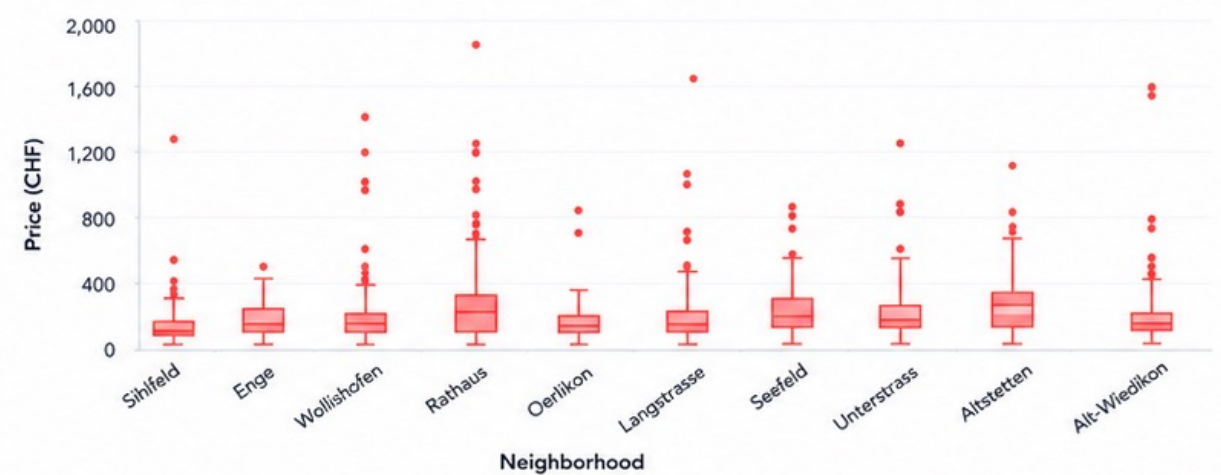


Widest Price Range
CHF 1,624.98+
Across Top 10

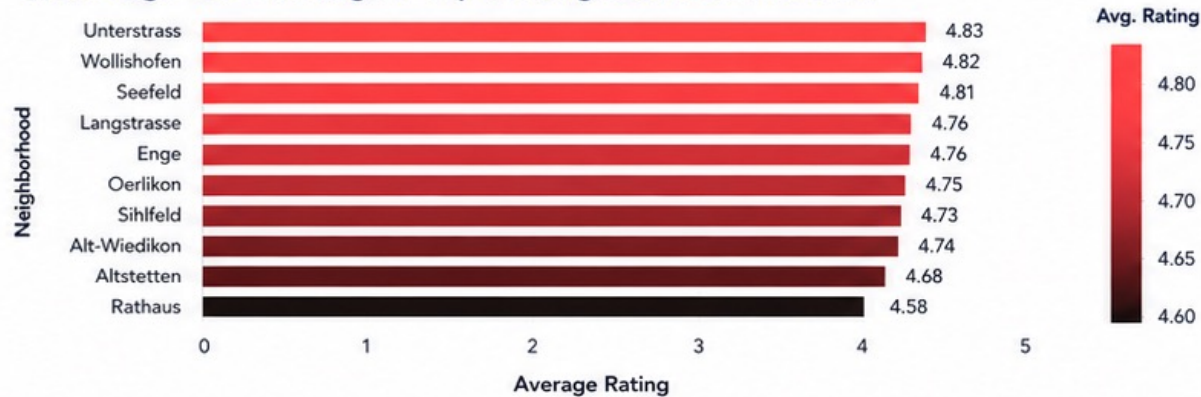
1. Average Airbnb Price for Top 10 Neighborhoods in Zurich



2. Price Distribution by for Top 10 Neighborhoods in Zurich (Mean + Variability)



3. Average Guest Rating for Top 10 Neighborhoods in Zurich



4. Average Number of Reviews by Top 10 Neighborhoods in Zurich (Demand Proxy)



KEY TAKEAWAYS

- Rathaus has the highest average price (CHF 267.86) and the highest average number of reviews (72.95).
- Unterstrass leads in guest satisfaction with the highest average rating (4.83).
- Price variability is high across all neighborhoods, with outliers exceeding CHF 1,600.
- Stronger ratings and higher review volume indicate stronger demand and market activity.



Airbnb Demand & Booking Insights – Zurich

Demand, Booking Behavior & Market Gaps Across Top 10 Neighborhoods



Highest Fully Booked Share

52.54%

Unterstrass



Largest Demand Gap

0.18

Oerlikon



Highest 30-Day Booking Rate

78.40%

Unterstrass



Highest 365-Day Booking Rate

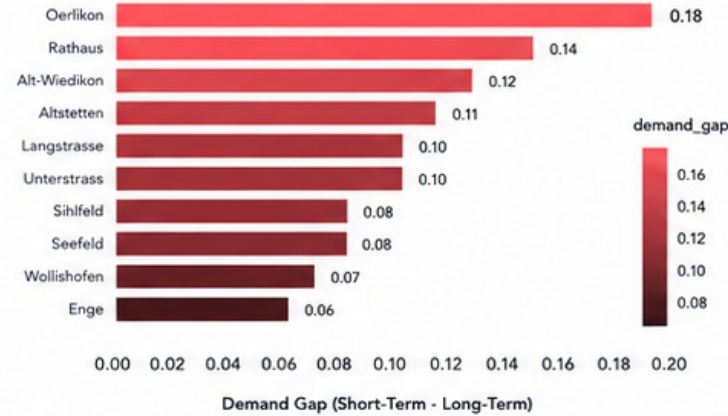
68.80%

Unterstrass

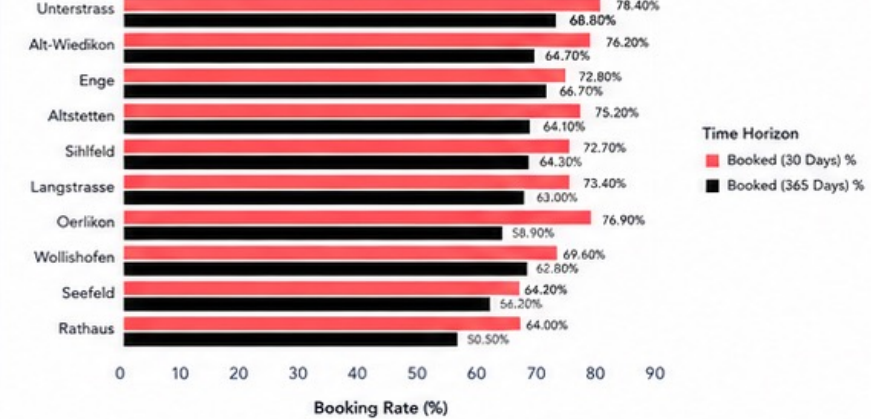
1. Share of Fully Booked Listings by Top 10 Neighborhoods in Zurich



2. Short-Term vs Long-Term Demand Gap by Neighborhood



3. Short-Term vs Long-Term Booking Rates by Neighborhood



KEY TAKEAWAYS

- Unterstrass leads in short-term demand with the highest fully booked share (52.54%) and the highest 30-day booking rate (78.40%).
- Oerlikon has the largest demand gap (0.18), indicating stronger short-term demand compared to long-term.
- Rathaus has the highest review volume (72.95), making it the most established and popular area.
- Neighborhoods like Alt-Wiedikon and Altstetten show strong booking rates across both short and long-term horizons.

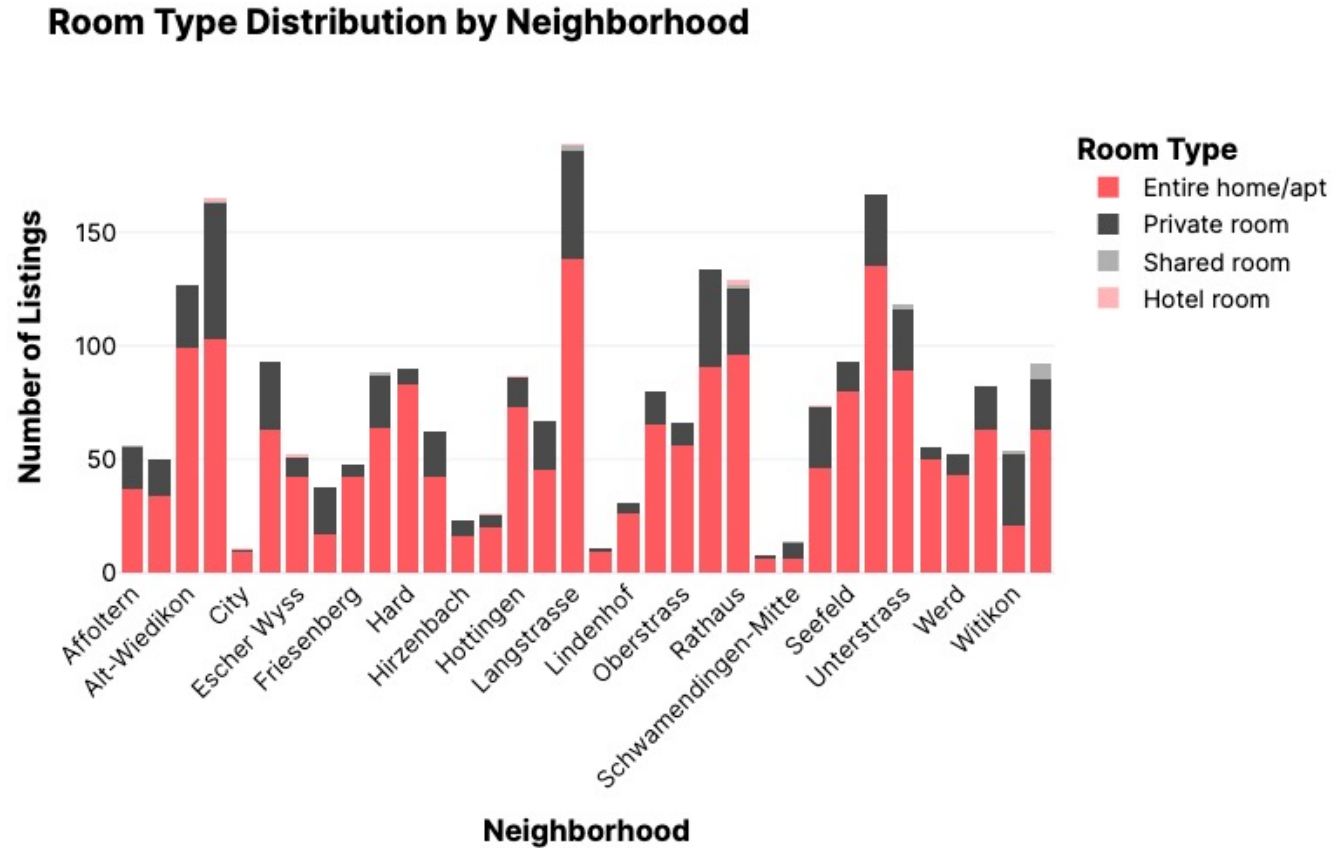


Data Visualizations

A closer look



Visualization: Room Type Distribution by Neighbourhood



What it shows

- Entire homes/apartments dominate listings across most neighborhoods
- Private/shared rooms appear more in select areas

Real-World Context

- Central areas (e.g., Langstrasse, Seefeld, Altstadt) → more entire-home rentals (tourist/business demand)
- Peripheral neighborhoods → more private/shared rooms (affordability-focused supply)

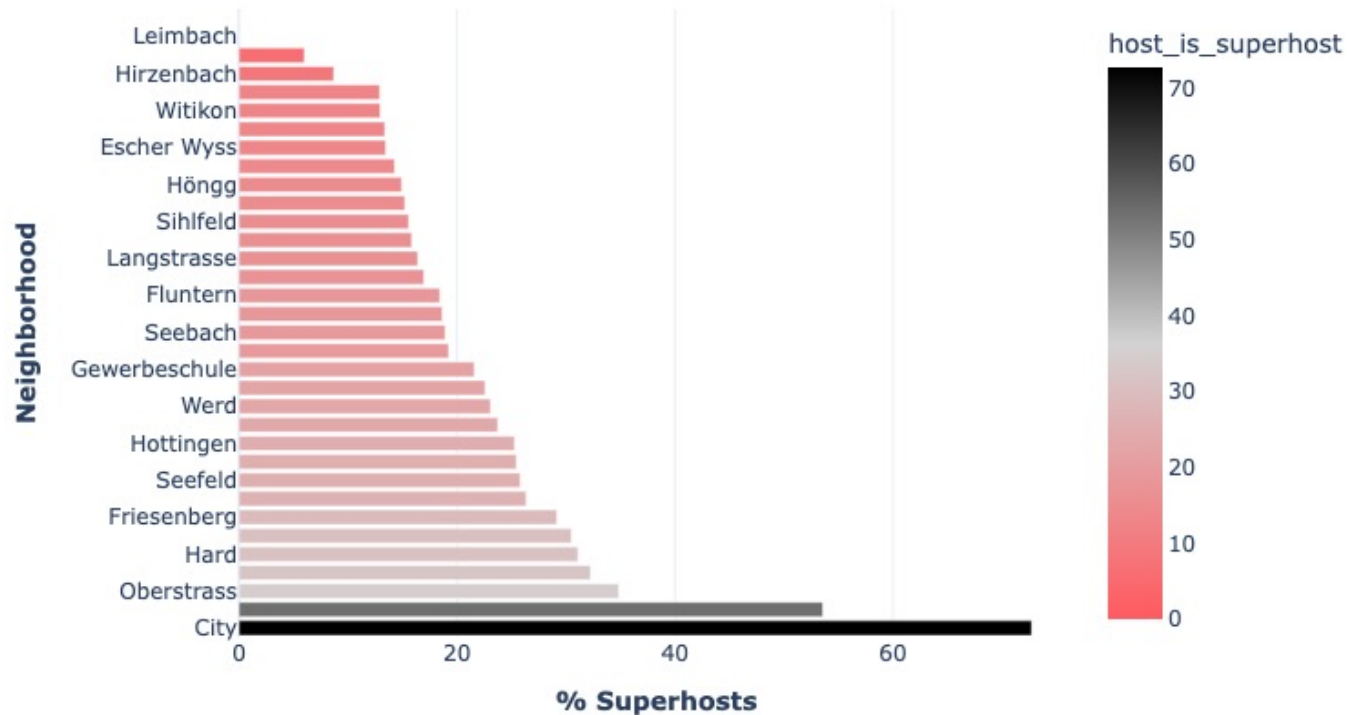
Insight

- Zurich Airbnb market is premium-skewed, with full-property rentals dominating high-demand zones



Visualization: Percentages of Superhosts by Neighbourhood

Percentage of Superhosts by Neighborhood



What it shows

- Significant variation in % of superhosts across neighborhoods
- Central and high-demand areas show higher superhost concentrations

Real-World Context (Zurich)

- Neighborhoods like:
 - City / Altstadt (Old Town) → tourism-heavy, high competition → hosts optimize quality → more superhosts
 - Seefeld → upscale, near Lake Zurich → professional hosting standards
- Outer residential areas → lower superhost rates due to more casual hosting

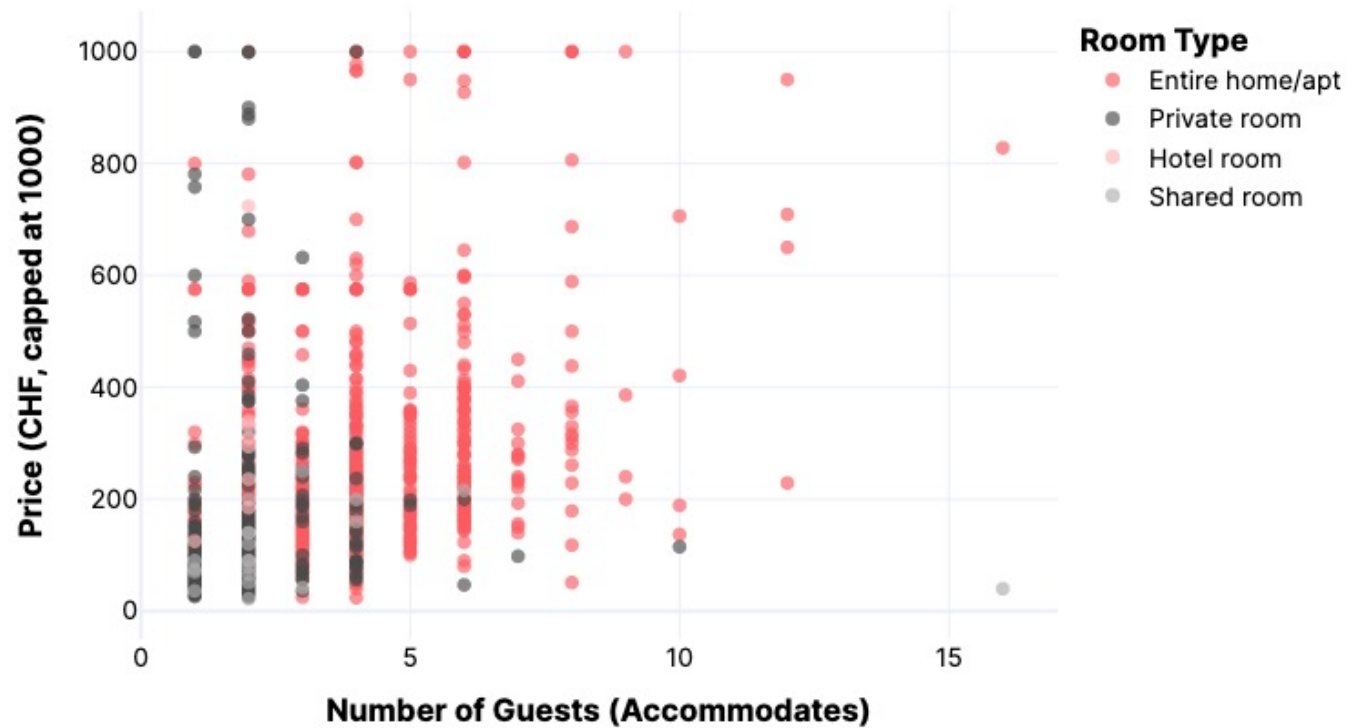
Insight

- High-demand, central neighborhoods push hosts toward higher service quality and professionalism



Visualization: Relationship between Listing Capacity & Price

Relationship Between Listing Capacity and Price in Zurich



What it shows

- Positive relationship: Listings that accommodate more guests tend to have higher prices
- High variability at each capacity level → pricing influenced by more than just size (location, amenities, property type)
- Entire homes dominate higher price ranges, especially for larger capacities

Insight

- Capacity increases price, but location + property quality drive premium pricing more strongly than size alone

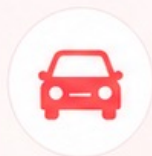


Machine Learning Models

KNN, Transformer, and Clustering Analysis

KNN Classification: Predicting Parking Availability

 **Goal:** Using k-nearest neighbors, predict whether a rental in your neighborhood will have some particular amenity.



AMENITY CHOSEN

Parking

TARGET PROPORTION

1	Has Parking	52.2%
0	No Parking	47.8%



PREDICTORS (FEATURES)



Accommodates



Bathrooms



Bedrooms



Beds



Price



MODEL CONFIGURATION

Algorithm

K-Nearest Neighbors (KNN)

Best K (Optimal)

7



MODEL PERFORMANCE

Accuracy

0.58

Confusion Matrix

	Predicted 0 (No Parking)	Predicted 1 (Has Parking)
Actual 0 (No Parking)	134	106
Actual 1 (Has Parking)	107	160



CLASSIFICATION REPORT

Class	Precision	Recall	F1-score	Support
0 (No Parking)	0.56	0.56	0.56	240
1 (Has Parking)	0.60	0.60	0.60	267
Accuracy	-	-	-	0.58
Macro Avg	0.58	0.58	0.58	507
Weighted Avg	0.58	0.58	0.58	507



BASELINE COMPARISON (NAIVE MODEL)

Majority Class Baseline

Always predicts the most frequent class from the training data.



Naive Accuracy

~0.53



The KNN model performs only slightly better than the naive baseline, indicating limited predictive power.



KEY TAKEAWAY

- Parking availability is not strongly explained by basic listing features alone.
- Other factors such as location, amenities, and property type may play a larger role.



airbnb Transformer Model: Predicting Price Quartile from Text

 **Goal:** Use a transformer model to build a classification model which predicts the price quartile (Q1–Q4) for a data record based on three descriptive text columns: **description**, **host_about**, and **amenities**.



MODEL OVERVIEW

Transformer Model
DistilBERT Base Uncased
(Hugging Face)

TASK



Multi-class classification
Predicting Price Quartile:
Q1 (lowest) → Q4 (highest)

INPUT TEXT SOURCES



description



host_about



amenities



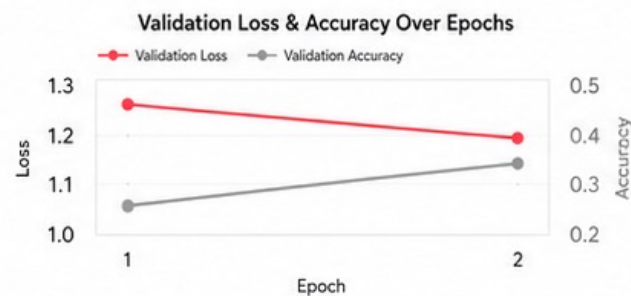
TRAINING RESULTS

Epoch	Training Loss	Validation Loss	Accuracy
1	No log	1.263177	0.408284
2	1.256944	1.212910	0.441815

BEST MODEL (AFTER 2 EPOCHS)



Validation Loss: **1.2129**
Validation Accuracy: **0.4418**



EVALUATION SUMMARY

eval_loss	1.2129
eval_accuracy	0.4418
eval_runtime (s)	4.8767
eval_samples_per_second	103.963
eval_steps_per_second	13.123
epoch	2.0

MODEL PERFORMANCE



The model achieves
44.18% accuracy on the
validation set.



EXAMPLE PREDICTION

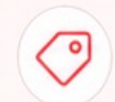
Input Text (Combined from 3 columns)

Beautiful luxury apartment in central Zurich with modern design, large balcony, and premium amenities.

Host is very responsive and provides excellent service.

Includes wifi, kitchen, washer, dryer, and parking.

Predicted Quartile



Q4_high



KEY TAKEAWAY

- ✓ The DistilBERT model can learn from rich textual information to predict price quartiles.
- ✓ There is room for improvement with more data, preprocessing, and hyperparameter tuning.





Goal: Perform k-means clustering to group rental units in Zurich based on key listing characteristics. Each observation in the dataset represents one rental unit.



FEATURES USED



Price (CHF)



Accommodates (Guests)



Number of Reviews



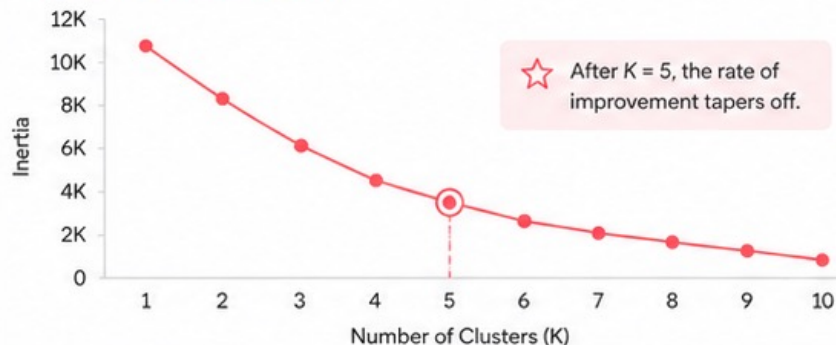
Review Scores Rating (1–5)



Booked 30 (Demand Proxy)

$\text{Booked_30} = 1 - \text{Availability_30} / 30$

ELBOW PLOT (K-MEANS)



Selected Number of Clusters: 5

Balances meaningful segmentation with interpretability.

CLUSTER SUMMARY (MEAN VALUES)

Cluster	Price (CHF)	Accommodates (Guests)	Number of Reviews	Review Scores Rating (1–5)	Booked 30 (Demand Proxy)
0	120.72	1.93	18.56	4.85	0.93
1	204.73	2.33	15.58	4.82	0.24
2	144.10	3.25	295.38	4.73	0.74
3	324.21	4.80	23.08	4.84	0.79
4	146.48	2.65	8.26	3.78	0.70

CLUSTER DESCRIPTIONS

Cluster	Name	Description
0	Budget High-Occupancy Listings	Small, lower-priced listings with very high short-term booking rates, suggesting strong demand for affordable stays.
1	Mid-Price Low-Demand Listings	Smaller listings with relatively higher prices but low booking rates, suggesting weaker demand or possible overpricing.
2	High-Reputation High-Demand Listings	Moderately priced listings with extremely high review counts and strong booking rates, indicating established, high-performing listings.
3	Premium Large-Capacity Listings	Larger, higher-priced listings with strong booking rates, representing premium or group-oriented rentals.
4	Lower-Rated Mid-Range Listings	Mid-range listings with lower review scores and fewer reviews, suggesting weaker guest satisfaction or less established performance.

KEY TAKEAWAYS



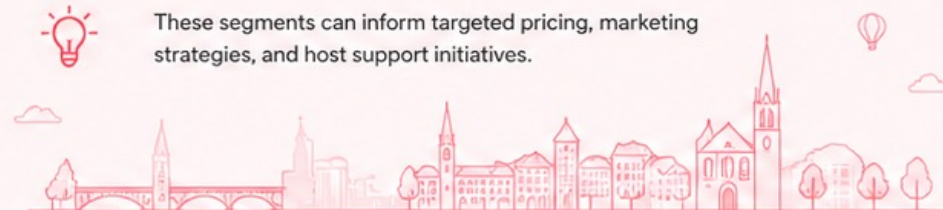
Five distinct clusters capture meaningful differences in price, capacity, popularity, and booking demand.



Listings range from budget, high-occupancy options to premium, large-capacity rentals.



These segments can inform targeted pricing, marketing strategies, and host support initiatives.



DATA PREPARATION & ENGINEERING



Handled missing values (median imputation)



Excluded 0 ratings (no reviews) and filled missing values



Created demand proxy:
 $\text{Booked_30} = 1 - \text{Availability_30} / 30$

Average Price and Demand Profile by Airbnb Cluster



Number of Listings in Each Airbnb Cluster



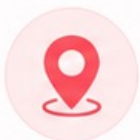
Average Price by Airbnb Cluster





Key Insights & Takeaways

Combining data analysis and machine learning to understand the **Zurich Airbnb market**.



Market Structure & Demand

- Strong central clustering of listings around the city center and lake – the core demand hub.
- Central areas command both higher prices and higher demand.
- Outer neighborhoods act as more affordable, lower-demand alternatives.



Pricing Dynamics

- Price generally increases with listing size (accommodates), but with high variability.
- Room type is a major driver – entire homes dominate higher price ranges.
- Significant price spread within similar listings suggests location, quality, and amenities matter.



Demand & Booking Behavior

- Short-term demand is strongest in key neighborhoods (e.g., Unterstrass).
- Some areas show demand gaps (e.g., Oerlikon) – opportunity for optimization.
- High review volume and ratings correlate with stronger booking performance.



Listing Characteristics

- Market is dominated by small (1–2 bedroom) listings, reflecting Zurich's dense, urban structure.
- Larger listings are rare and act as premium or niche offerings.
- Entire homes dominate central areas, while shared/private rooms are more dispersed.



Machine Learning Insights

- KNN model showed limited predictive power – basic features alone don't explain amenities well.
- Transformer model captured meaningful signal from text – unstructured data is valuable.
- Clustering revealed 5 distinct market segments, from budget high-demand to premium listings.



Strategic Takeaways

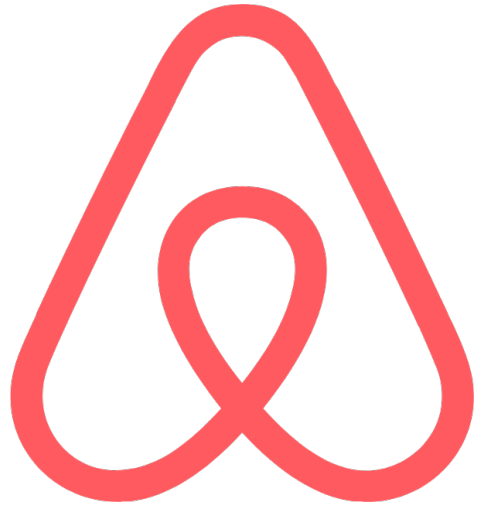
- Location is the strongest driver of both price and demand.
- Market segmentation enables targeted pricing and positioning strategies.
- Combining structured + unstructured data improves predictive understanding of listings.



Overall Takeaway

The Zurich Airbnb market is driven by location, constrained by housing structure, and best understood through a combination of data-driven modeling and segmentation.





Thank you for
listening!

